

Statement of Work (SOW)

Project Title

ChargeWise AI: Intelligent EV Charging Infrastructure Planning Platform

1. Executive Summary

Nigeria's electric vehicle (EV) ecosystem is growing, but infrastructure deployment remains largely reactive due to fragmented mobility and demand data. Governments, investors, and charging operators lack the tools needed to identify optimal locations for charging stations and forecast future demand.

ChargeWise AI is a cloud-native decision-support platform that leverages AI and AWS services to analyze mobility patterns, population density, commercial activity, and existing charging infrastructure. The platform provides intelligent recommendations on where EV charging stations should be deployed to maximize utilization, improve accessibility, and support sustainable transportation growth.

2. Project Overview

Problem Statement

The absence of a centralized EV infrastructure planning system makes it difficult for stakeholders to determine:

Where future charging demand will emerge.

Which locations offer the highest utilization potential.

How infrastructure investments should be prioritized.

Proposed Solution

ChargeWise AI aggregates and processes multiple datasets to generate:

EV demand forecasts

Infrastructure gap analysis

Charging station recommendations



Investment and utilization insights

The platform provides a visual dashboard that helps stakeholders make data-driven infrastructure decisions.

3. Business Case

Current Challenges

Limited visibility into EV demand trends.

Inefficient allocation of charging infrastructure investments.

Lack of predictive planning tools.

Increased risk for investors entering the EV market.

Business Value

ChargeWise AI enables:

- **Governments**

Better urban mobility planning.

Evidence-based infrastructure development.

- **Investors**

Reduced investment risk.

Improved return on infrastructure projects.

- **Charging Operators**

Strategic site selection.

Increased charging station utilization.



Expected Impact

Faster EV adoption.

Better charging accessibility.

Improved infrastructure utilization.

Data-driven investment decisions.

4. High-Level Technical Architecture

Data Sources

Population Density Data

Traffic Density Data

Commercial Activity Data

Existing Charging Station Data

AWS Components

Amazon S3 (Data Lake)

AWS Glue (Data Processing)

Amazon SageMaker (AI Demand Prediction)

Amazon DynamoDB (Data Storage)

AWS Lambda (Business Logic)

Amazon API Gateway (API Management)

Amazon CloudFront (Content Delivery)

Amazon CloudWatch (Monitoring)



User Interface

Web dashboard providing:

Demand heatmaps

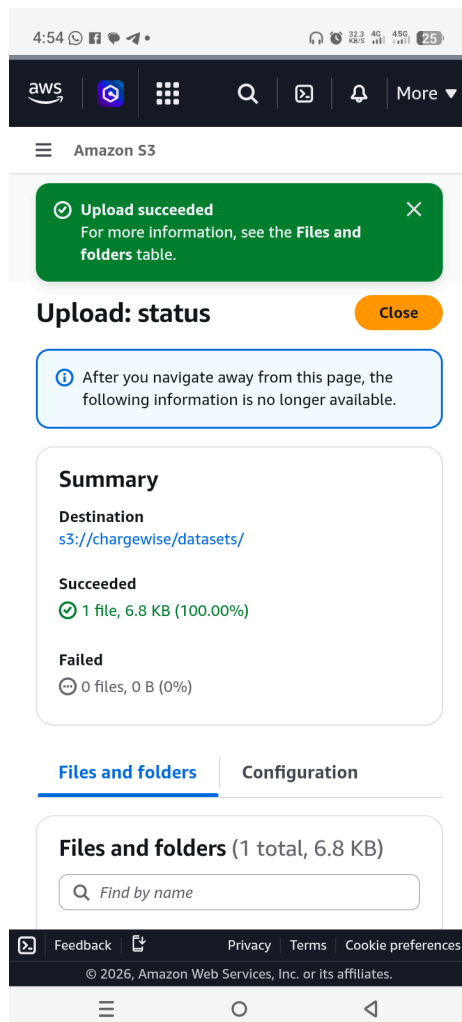
Infrastructure recommendations

Analytics and reporting

AWS ARCHITECTURE



S3 bucket



Lambda function

aws

United States (N. Virginia)

Taiwo (203012547348)

Lambda > Functions > chargewise-recommendation-engine

chargewise-recommendation-engine

Throttle Copy ARN Actions

Function overview

Diagram Template

chargewise-recommendation-engine

Layers (0)

API Gateway

+ Add trigger

+ Add destination

Function ARN

arn:aws:lambda:us-east-1:203012547348:function:chargewise-recommendation-engine

Description

-

Last modified

6 minutes ago

Code Test Monitor Configuration Aliases Versions

Code source

Open in Visual Studio Code Update

EXPLORER

CHARGEWISE-RECOMMENDATION-ENGINE

lambda_function.py

lambda_function.py

3

4

5

13

14

15

16

17

18

PROBLEMS OUTPUT CODE REFERENCE LOG TERMINAL Tasks

Ln 14, Col 148 Spaces: 8 UTF-8 LF Python Lambda Layout: us

DEPLOY

Deploy (Ctrl+Shift+U)

Test (Ctrl+Shift+I)

TEST EVENTS [SELECTED: TEST-EVENT]

Create new test event

Shareable saved events

test-event

ENVIRONMENT VARIABLES

Lambda Deployed 0 0 Amazon Q

Code properties

Package size

340 byte

SHA256 hash

Ukw9wOxIPmMIE5i4bxuQrzMXgWqbNi0IB3o0Bylnenc=

Last modified

6 minutes ago

Runtime settings

Edit Edit runtime management configuration

Runtime

Python 3.14

Handler

Info

lambda_function.lambda_handler

Architecture

Info

x86_64

Runtime management configuration

Runtime version ARN

Info

arn:aws:lambda:us-east-1:runtime:28933846e9c8fabbfc0f38cf5337c13c4e46a0f226f1f0559d98c271ed4b22d8

Update runtime version

Info

Auto

CloudShell Feedback Console Mobile App

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API gateway route

The screenshot displays the AWS API Gateway console interface. The left-hand navigation pane is expanded to show the 'Routes' section under the 'API Gateway' heading. The main content area is titled 'Routes for chargewise-recommendation-api' and includes a 'Deploy' button. Below this, a search bar is present, and a list of routes is shown, with the selected route being 'GET /recommendations'. The 'Route details' section for this route is visible, showing the method 'GET', the path '/recommendations', and the ARN 'arn:aws:apigateway:us-east-1::apis/06mru8m2b/routes/4yor7k3'. The 'Authorization' section indicates that no authorizer is currently attached to the route, with an 'Attach authorization' button. The 'Integration' section shows that the route is integrated with a backend resource named 'vnot1ls', with a 'Configure' button available.

API Gateway

APIs

Custom domain names

Domain name access associations

VPC links

API: chargewise-reco... (06mru8m2b)

▼ Develop

Routes

Authorization

Integrations

CORS

Reimport

Export

▼ Deploy

Stages

▼ Monitor

Metrics

Logging

▼ Protect

Throttling

Routes

Stage: - ▼ Deploy

Routes for chargewise-recommendation-api Create

Search

▼ /recommendations

GET

Route details Delete Edit

GET /recommendations (ID: 4yor7k3)

ARN

arn:aws:apigateway:us-east-1::apis/06mru8m2b/routes/4yor7k3

Authorization

Authorizers protect your API against unauthorized requests. Routes with no authorization attached are open.

No authorizer attached to this route. Attach authorization

Integration

The integration is the backend resource that this route calls when it receives a request.

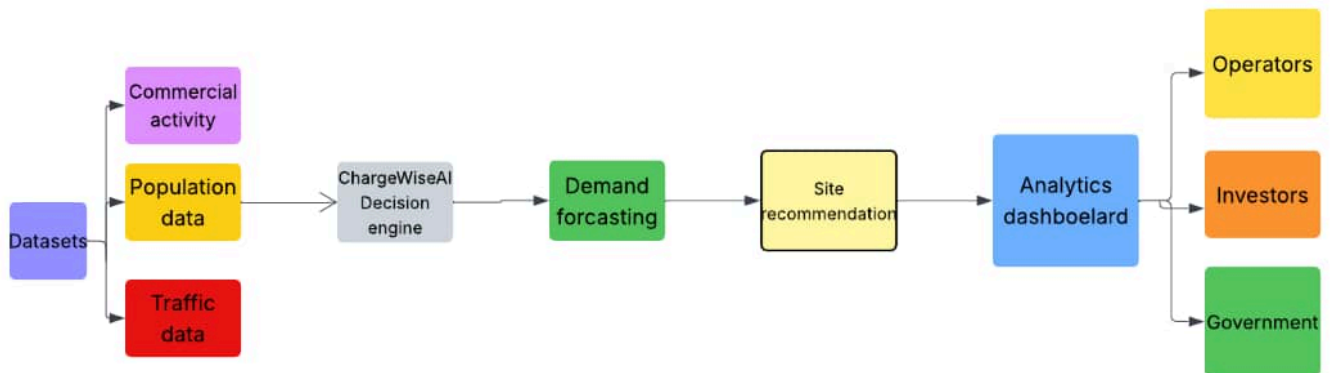
vnot1ls Configure



JSON response

```
06nwru8m2b.execute-api.us-east-1.a...  
06nwru8m2b.execute-api.us-east-1.amazonaws.com  
[{"location": "Lekki", "score": 92, "rank": 1}, {"location": "Ikeja", "score": 88, "rank": 2}]
```

BUSINESS ARCHITECTURE



5. Functional Requirements

FR-1 Data Ingestion

The system shall collect and store mobility and infrastructure datasets.

FR-2 Data Processing

The system shall clean, transform, and prepare data for analysis.

FR-3 Demand Prediction

The system shall estimate future EV charging demand for selected locations.

FR-4 Location Recommendation

The system shall rank locations based on projected charging demand.

FR-5 Interactive Dashboard

The system shall display:

Demand heatmaps

Recommended charging locations

Infrastructure coverage analysis

FR-6 Reporting

The system shall generate planning insights for stakeholders.

6. Technical Specifications

Frontend

React.js

Tailwind CSS

Chart.js / Recharts

Google Maps API or Leaflet

Backend

- AWS Lambda
- Amazon API Gateway
- Data Storage
- Amazon S3
- Amazon DynamoDB



AI/Analytics

- Amazon SageMaker
- Python
- Pandas
- Scikit-learn
- Monitoring
- Amazon CloudWatch

7. Project Phases & Deliverables

Phase ActivityDeliverable

Phase 1	Problem Analysis	Requirements Definition
Phase 2	Architecture Design	AWS Architecture Diagram
Phase 3	Data Preparation	Processed Sample Dataset
Phase 4	AI Development	Demand Prediction Model
Phase 5	Dashboard Development	Interactive Web Dashboard
Phase 6	Testing & Demo	Functional Prototype
Phase 7	Documentation	SOW & Presentation Materials



8. Success Criteria

- The project will be considered successful if it:
- Technical Success
- Successfully processes mobility and infrastructure data.
- Generates demand forecasts.
- Recommends suitable charging station locations.
- User Success
- Stakeholders can visualize infrastructure gaps.
- Users can identify high-potential investment locations.

Business Success

- Demonstrates improved infrastructure planning capability.
- Supports data-driven decision making.

9. Risk Management

Risk	Impact	Mitigation
Limited real-world data	Medium	Use publicly available and simulated datasets
Time constraints	High	Focus on MVP features
AI model accuracy	Medium	Use explainable scoring model
AWS service configuration issues	Medium	Use managed services and simplified architecture



Integration challenges	Medium	Develop modules independently and integrate early

10. Conclusion

ChargeWise AI addresses a critical challenge in Nigeria's emerging EV ecosystem by providing an intelligent platform for infrastructure planning. By combining AI-driven forecasting with AWS cloud technologies, the solution enables governments, investors, and operators to make informed decisions about EV charging station deployment, accelerating sustainable transportation development across the country.

